

SEUNFUNMI ADEGOKE

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PROFESSIONAL SUMMARY

Senior AI Engineer with experience building production-grade LLM systems for agentic AI, RAG, document intelligence, model routing, and AI evaluation in business-critical workflows. Architected high-throughput AI agents, due-diligence QA systems, retrieval pipelines, and tool-calling infrastructure serving tens of thousands of daily users and millions of model requests. Strong backend engineering foundation in Python, FastAPI, async systems, Kafka, Kubernetes, vector databases, LangGraph, LlamaIndex, Langfuse, LiteLLM, and cloud AI platforms, with a track record of improving reliability, latency, retrieval quality, and model/tool-use accuracy in production environments.

SKILLS

- **Programming Languages:** Python, TypeScript, JavaScript, React, Bash
- **LLM Systems & Agentic AI:** AI agents, agent orchestration, multi-step workflows, RAG, tool calling, function calling, long-term memory, prompt engineering, context engineering, structured output validation, LLM evaluation, model safety
- **Agent & RAG Frameworks:** LangGraph, LangChain, LlamaIndex, DSPy, OpenAI Agents SDK, PydanticAI, Model Context Protocol, CrewAI
- **Backend & APIs:** FastAPI, Django, Flask, REST APIs, asyncio, async pipelines, microservices
- **Data, Search & Vector Infrastructure:** PostgreSQL, pgvector, Azure AI Search, Kafka, MySQL, MongoDB, DynamoDB, Neo4j, Chroma, Qdrant, Weaviate, Pinecone
- **LLMOps, Observability & Model Routing:** Langfuse, LangSmith, LiteLLM, Logfire, Lunary, Grafana, Datadog, evaluation pipelines, model gateways, provider failover, guardrails
- **Model Providers & Serving:** OpenAI, Anthropic, AWS Bedrock, Azure AI Foundry, OpenRouter, Groq, Meta Llama, DeepSeek, Hugging Face, vLLM, Ollama
- **Cloud, DevOps & Reliability:** AWS, Google Cloud, Azure, Docker, Kubernetes, Terraform, CI/CD

WORK EXPERIENCE

Salus Optima

Nov. 2025 – Present

Senior AI Engineer

- Architected Salus' LangGraph QA agent for financial due diligence, enabling users to ask complex investment and company-analysis questions over uploaded documents and receive grounded answers. Built the multi-node state machine, conditional routing, memory checkpointing, and RAG answer-synthesis workflow, scaling the system to approximately 45,000 questions/day with 89% answer accuracy.
- Built the upload-time RAG ingestion pipeline for Excel, PowerPoint, email, image, Word, and PDF files, turning heterogeneous diligence materials into searchable, LLM-ready knowledge sources. Implemented document-to-markdown conversion, Azure AI Search indexing via LlamaIndex, DRM detection, malformed-document repair, scoped retries, and indexing timeouts, processing 120 documents/minute at a 94% ingestion success rate.
- Implemented a two-tier answering strategy that served cached vector-indexed answers for repeat questions and used multi-judge LLM verification for quality control. Reduced average response time for repeat questions from 12 seconds to under 4 seconds while preserving answer quality.
- Engineered self-correcting LangGraph tools with state injection, including an LLM-powered chart/table generator that render-validates its own output and retries with structured error feedback, supporting AI-generated analysis and reporting workflows.

- Owned AI platform reliability and model infrastructure, maintaining self-hosted Langfuse on Kubernetes at 99.9% uptime and unifying Anthropic, AWS Bedrock, and Azure AI Foundry behind a provider-agnostic LLM interface for observable production AI workflows.

Replika

Feb. 2025 – Nov. 2025

AI Engineer

- Architected Replika's core AI agent, a streaming plan-then-execute system that decomposed user requests into dependency-aware tool calls and executed them in parallel. Built the underlying async scheduler with 16-way concurrency, per-tool timeouts, and failure isolation, enabling graceful degradation and reliable operation across 1.3 million conversations/day.
- Built a fault-tolerant agent runtime for live model outputs, repairing malformed JSON, validating structured tool calls from the token stream, and dispatching executable tools before planning completed. Increased valid-plan execution to 89% and reduced agent failures by 7%.
- Built an event-sourced long-term memory service that converted chat, email, calendar, and web data into structured user memory for personalization and retrieval. Implemented Kafka-based reducers, concurrent LLM fact extraction, deterministic deduplication, lineage-tracked deletes, and pgvector retrieval, improving retrieval relevance by 38%.
- Built a multi-tenant data ingestion platform that connected Gmail, Google Calendar, and web data to Replika's AI systems, enabling reliable personalization data sync across users. Implemented pluggable handlers, OAuth2 lifecycle management, backfills, incremental sync, and Gmail deduplication, reducing redundant email ingestion by 23%.
- Centralized model access through a LiteLLM gateway routing 42 model aliases across OpenAI, OpenRouter, and Groq, with provider failover, cost-aware routing, and capability-aware dispatch. Provided all services with a single observable model endpoint serving 5.3 million requests/day.

Turing

Jan. 2023 – Feb. 2025

AI Engineer

- Built ChatGPT plugin and tool-calling integrations end-to-end, enabling models to call live third-party services through compliant manifests, OpenAPI specifications, and backend APIs. Improved function calling success on plugin tasks by approximately 25%.
- Built an Employee Handbook RAG system serving 15,000+ staff, enabling employees to answer HR policy, benefits, onboarding, and compliance questions with cited sources from authoritative internal documents. Designed the RAG evaluation framework across retrieval precision, citation faithfulness, answer groundedness, refusal behavior, and policy coverage, reducing HR support tickets by 35%.
- Built an evaluation benchmark for ChatGPT plugin and tool use, scoring tool selection, argument construction, citation quality, structured output validity, and rubric adherence across hundreds of scenarios. Surfaced recurring model failure modes and improved plugin reliability by approximately 20%.

Sendme.ng

Jan. 2022 – Jan 2023

Software Engineer

- Architected and deployed a Dialogflow-based conversational AI chatbot for order.sendme.ng, automating customer support and order-tracking workflows across web and mobile channels. Designed intent models, fulfillment webhooks, and NLP flows for order status, delivery updates, and customer issue resolution.

E-LOPA

June 2021 – Dec. 2021

Software Engineer

- Built the backend for a Bowtie risk assessment platform for chemical process plants using Python, Django, and PostgreSQL, supporting structured process-safety analysis for small and medium-scale operations.

Mentorpaired

Dec. 2020 – June 2021

Software Engineer

- Designed backend architecture and database models with a team of 6 engineers for a **peer-to-peer book loan platform** using Python, Flask, and React, including custom authentication and authorization flows.